



Operations & Resource Optimization

19%

of calls went unanswered.
We set out to find why.

*A data driven analysis of 5000 calls across 8 agents – from
Raw data to dashboard.*



Project Overview

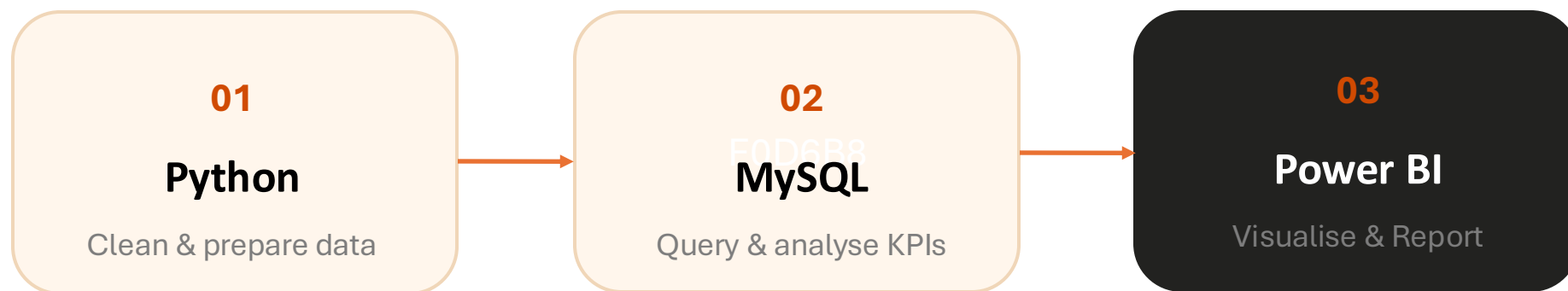
ABOUT THIS PROJECT

An end-to-end analysis turning 5,000 raw call records into operational insight

This project analyses **Q1 2021 Call Centre operations** from PwC's public dataset to identify why nearly one in five calls went unanswered - and where the operation can improve.

- ▶ 5,000 calls · January- March 2021
- ▶ 8 agents across 5 call topics
- ▶ Source: PwC Call Centre Trends Dataset

How It Was Built – 3 Stage Workflow



◆ **Outcome:** A 2-page interactive dashboard with cross-validated KPIs and 3 actionable recommendations for management.



Key Findings - The Headline Numbers

FINDING 01

The team resolves nearly 9 in 10 calls it answers - but answers only 4 in 5

Five KPIs summarizing 5,000 calls across Q1 2021

5,000

Total Calls

81%

Answer Rate

89.9%

Resolution Rate

3.40

Avg Satisfaction

67.5s

Avg Speed

Strong resolution, but a costly **answer-rate gap**.

When a call is answered, it has resolved 89.9% of the time - a genuinely strong service level. The problem is not quality; it is capacity. Nearly one in five calls is never answered, representing lost customers and missed resolution opportunities before an agent ever engages.

946

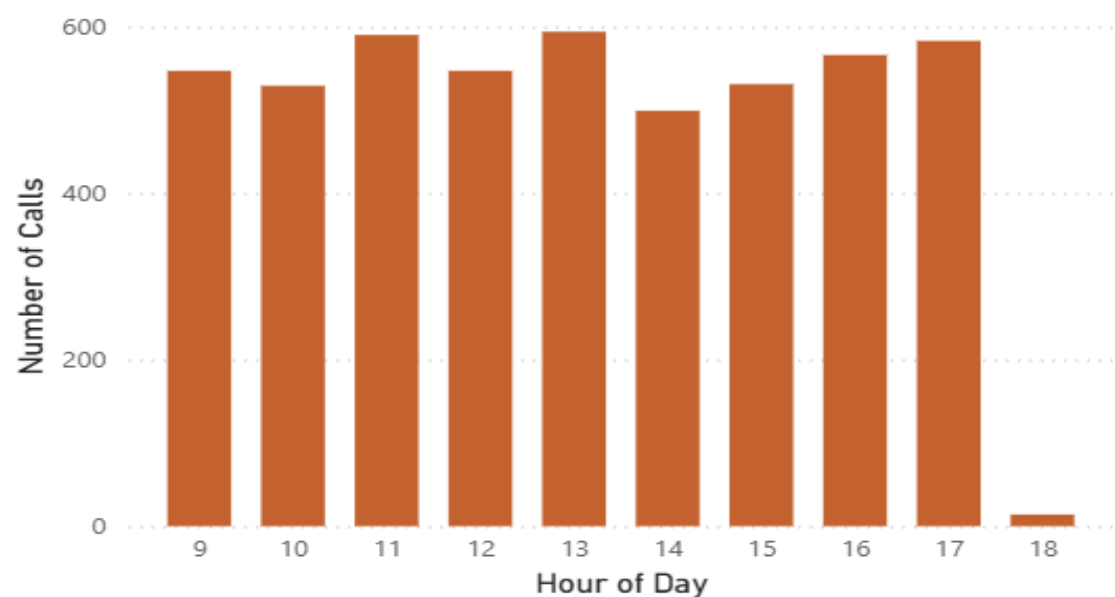
calls went unanswered
(19% of all calls)



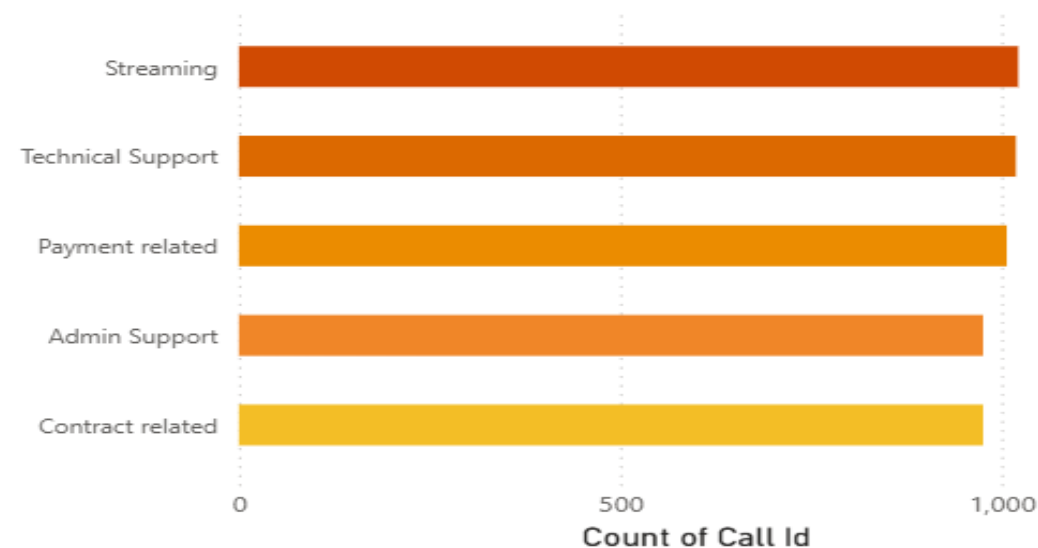
Key Findings - Where the Gap Lives

FINDING 02

The unanswered-call gap isn't a rush hour - it's constant across every hour and topic



Call Volume by Topic

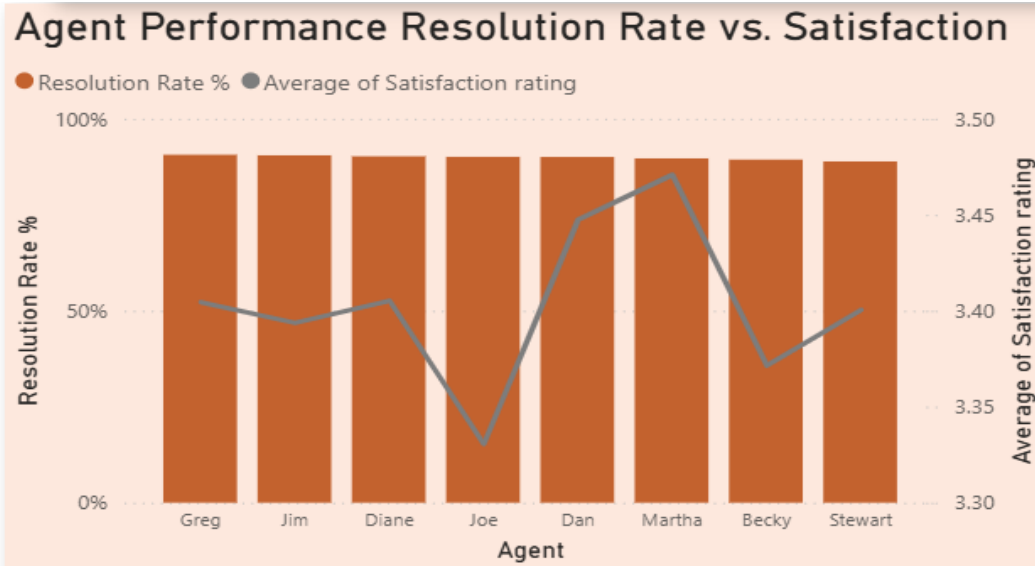


◆ **What this means:** Call volume peaks around **11am & 1pm**, yet the answer rate holds near **81% all day** - and all five topics carry near-equal volume. Because the gap appears everywhere equally, it points to a baseline capacity shortfall, not a peak-hour or single-topic bottleneck.

Key Findings - The Hidden Specialists

FINDING 03

Every agent looks alike on resolution - until topic-level data reveals clear strengths and blind spots



Avg Satisfaction by Topic

Agent	Admin Support	Contract related	Payment related	Streaming	Technical Support	Total
Becky	3.44	3.29	3.48	3.32	3.31	3.37
Dan	3.51	3.26	3.42	3.45	3.60	3.45
Diane	3.33	3.49	3.51	3.34	3.38	3.41
Greg	3.35	3.44	3.31	3.44	3.49	3.40
Jim	3.42	3.36	3.27	3.51	3.41	3.39
Joe	3.49	3.42	3.26	3.22	3.29	3.33
Martha	3.47	3.38	3.46	3.59	3.45	3.47
Stewart	3.43	3.40	3.45	3.35	3.37	3.40
Total	3.43	3.38	3.40	3.40	3.41	3.40

◆ **What this means:** Resolution rates sit within a tight **1.7-point band** (88.9–90.6%), suggesting agents are interchangeable. But satisfaction by topic tells a different story: **Dan leads Technical Support** (3.60) yet lags on Contract calls (3.26). This hidden specialisation is invisible in aggregate metrics - and it's a lever for smarter call routing.

Recommendations

FROM INSIGHT TO ACTION

Three moves to close the answer-rate gap - without sacrificing quality

1 Increase baseline capacity

Add front-line headcount or flex scheduling to lift the answer rate. The shortfall is steady all day, so a consistent capacity increase - not peak-only staffing - is the fix.

ADDRESSES

Finding 01 & 02 - 19% unanswered, constant across hours

Impact: Directly recovers a share of 946 lost calls

2 Skills-based call routing

Route calls to agents who score highest on that topic - e.g. Technical Support to Dan, Contract calls to Diane. Lifts satisfaction using existing staff strengths.

ADDRESSES

Finding 03 - hidden topic specialisation

Impact: Higher satisfaction at zero added headcount

3 Targeted topic coaching

Pair each agent's strongest topic with their weakest for peer coaching - e.g. Dan mentors on Technical Support while improving on Contract calls.

ADDRESSES

Finding 03 - per-agent blind spots

Impact: Narrows the satisfaction spread over time



The Interactive Dashboard

BUILT IN POWER BI

A 2-page interactive dashboard that lets managers explore the data themselves



◆ **Interactive slicers**
Filter any view by agent

◆ **Custom DAX measures**
Answer & resolution rates

◆ **Drill-down detail**
Agent x topic heatmap

◆ **Cross-validated**
Matches Python & SQL



DATA ANALYSIS PORTFOLIO PROJECT
THANK YOU

From raw data to decision-ready insight.

Amisha Gadekar - Aspiring Business Analyst

This project demonstrates an **end-to-end analytics workflow** - cleaning and preparing data in Python, querying KPIs in SQL, and building an interactive Power BI dashboard - to turn 5,000 raw call records into **three evidence-based recommendations** for management.

TOOLS USED

Python

MySQL

Power BI

DAX

SKILLS DEMONSTRATED

Data cleaning

SQL analysis

Data visualisation

Dashboard design

Business storytelling